Multivariate Calculus Lecture Notes
Section 15.1 – Integration in Two Variables

I. Definition of a double integral
   a. Problem: Estimate the volume of the solid that lies above the square $R = [0, 2] \times [0, 2]$ and below the elliptic paraboloid $z = 16 - x^2 - 2y^2$.
      i. Graphically:

      ii. You can imagine estimating the volume of the solid by boxes.
      iii. The more boxes, the better the approximation of the volume.
   b. In fact, if we let the size of the rectangles used to form the base of the boxes become infinitesimally small then we will have the exact volume of the solid.
      i. The region R, graphically:

      ii. Let ΔA_{ij} be the area of rectangle ij in partition P, and let P_{ij} be a representative point in rectangle ij.
      iii. Then the volume of box ij is $f(P_{ij})\Delta A_{ij}$.
      iv. The volume of the entire solid is approximately $\displaystyle\sum_{i=1}^{m}\sum_{j=1}^{n} f(P_{ij})\Delta A_{ij}$.
      v. Let $\|P\|$ denote the *norm* of the partition P, i.e., the largest height or width of any rectangle in the partition.
      vi. Then $V = \displaystyle\lim_{\|P\|\to 0}\sum_{i=1}^{m}\sum_{j=1}^{n} f(P_{ij})\Delta A_{ij}$.
      vii. This limit is defined to be the double integral of f over the region R:

$$\iint_R f(x, y)\, dA = \lim_{\|P\|\to 0}\sum_{i=1}^{m}\sum_{j=1}^{n} f(P_{ij})\Delta A_{ij}.$$

II. Properties of double integrals
   a. If $f(x, y)$ is continuous on a rectangle R then $f(x, y)$ is integrable over R.
   b. (Linearity Property.) If f and g are integrable over R then

$$\iint_R (f(x, y) + g(x, y))\, dA = \iint_R f(x, y)\, dA + \iint_R g(x, y)\, dA.$$

   c. If C is a constant, then $\displaystyle\iint_R Cf(x, y)\, dA = C \iint_R f(x, y)\, dA$.
   d. $\displaystyle\iint_R dA$ gives the area of rectangle R.

III. Evaluating a double integral
   a. How do we evaluate something like this, though?
   b. One approach to finding volume is to work with cross-sections.

i. Graphically:

ii. If $CSA(x)$ denotes the area of the cross-section for fixed x, then the volume is $\int_0^2 CSA(x)dx$.

iii. And $CSA(x) = \int_0^2 (16 - x^2 - 2y^2)dy$.

iv. Thus the volume of the solid is $\int_0^2 \int_0^2 (16 - x^2 - 2y^2)dydx$.

v. This is

$$\int_0^2 \left[16y - x^2y - \frac{2}{3}y^3\right]_0^2 dx = \int_0^2 \left(32 - 2x^2 - \frac{16}{3} - (0 - 0 - 0)\right) dx$$
$$= \int_0^2 \left(\frac{80}{3} - 2x^2\right) dx = \left[\frac{80}{3}x - \frac{2}{3}x^3\right]_0^2 = \frac{160}{3} - \frac{16}{3} - 0 = 48.$$

vi. It's embedded in the calculation, but $CSA(x) = 80/3 - 2x^2$.

c. We could have done this integration with y cross-sections first, and we would have gotten the same result.

d. Fubini's Theorem: If $f(x, y)$ is continuous on $R = [a, b] \times [c, d]$ then

$$\iint_R f(x, y)dA = \int_a^b \int_c^d f(x, y)dy\,dx = \int_c^d \int_a^b f(x, y)dx\,dy.$$

e. Example: $\iint_R xy\cos(y)dA$, where $R = [-1, 1] \times [0, \pi]$.

i. Which variable is easier to integrate with respect to first?

ii. If we integrate with respect to x first this is $\int_0^\pi \int_{-1}^1 xy\cos(y)\,dx\,dy$.

iii. Notice that $y\cos y$ is a constant with respect to the inner integral, and so we can factor it out: $\int_0^\pi y\cos y \left(\int_{-1}^1 x\,dx\right) dy$.

iv. Now the entire x integral is a constant with respect to y (it doesn't have any y values in it), and so it can be factored out to obtain $\left(\int_{-1}^1 x\,dx\right)\left(\int_0^\pi y\cos y\,dy\right)$.

v. The x integral is easy: $\int_{-1}^1 x\,dx = \frac{1}{2}x^2\Big|_{-1}^1 = \frac{1}{2}(1 - 1) = 0$.

vi. Thus the value of the integral is 0.

vii. Let's do the y integral for kicks.

1. We'll have to do it by parts.
   - $u = y$, $dv = \cos y\,dy$
   - Thus $du = dy$, $v = \sin y$
2. Thus $\int_0^\pi y\cos y\,dy = y\sin y\Big|_0^\pi - \int_0^\pi \sin y\,dy = 0 + \cos y\Big|_0^\pi = -1 - 1 = -2$.

f. Comments

i. If the bounds are constant, and the integrand can be factored into a function solely of x and a function solely of y, then the double integral can be factored into the product of two separate integrals in x and y.
ii. Thanks to Fubini's Theorem, you get to pick the order in which you do the integration.
    1. Try to pick an order that makes the integration easier.
    2. This will be more important when we start looking at regions with non-constant bounds.

Homework: Section 15.1 #6, 9 (do without calculus), 20, 23, 26, 29, 32, 34, 41, 47